

Business Data Management Capstone Project

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Midterm Submission

Title: Analyzing the Revenue Trend & Sale Volume and exploring possibilities of boosting the existing revenue generation of a stationery shop

Executive Summary

I am Soham Bhattacharya. As mentioned in my proposal, I had approached a 25-year old local stationery shop named 'Nandi Gift House'. It sells stationery items primarily along with cosmetic goods and some food items. Mr. Shantanu Nandi is the owner of the shop. Currently, it is being maintained and taken care of by his sons; primarily his eldest son, Mr. Abhishek Nandi. It has an appreciable reputation in the locality for the flawless services they have been providing so long. Mr. Nandi provided me with the raw data on 26th December, 2022. He usually maintains the register for the business dealings and sales. Data for a period of 4 months (August to November) were provided. Recently, they started to maintain their daily data in digital format i.e. in Excel Spreadsheet. So, the data received was in an Excel file. A short interaction video was also shot with Mr. Abhishek.

Initially, data cleaning was required since the raw data received was not properly formatted or had multiple errors in text. Data parsing was also done.

Next, according to the objective, the data was examined to analyze the different scopes to conclude solutions for the problems using different analysis techniques and tools from Excel.

Proof of originality of the data

Link of the raw data and its corresponding analysis : [Nandy Gift House - 26 December 2022.xlsx](#)

Interaction video link :
https://drive.google.com/file/d/1ClfPtmtLMS5AGrwsKPv6F70IxcEYTbSA/view?usp=share_link

Interactive Video Transcription

Mr. Abhishek Nandi is the speaker in the video.

Abhishek Nandi: Greetings! I'm Abhishek Nandi. Our shop was inaugurated in 1992 by my grandfather, Mr. Birendra Kumar Nandi. It has been two years since my father passed the baton to me. Along with my two brothers, I am currently in charge of all sorts of business tasks. The shop initially begun by selling gift items and some decorative items for occasional purposes. Eventually down the lane, the business expanded and as a result the stock volume increased.

Question: So during the lockdown period, how did you tackle and face the difficulties that arose against the business?

Abhishek Nandi: In the initial stages of lockdown, it was difficult to cope up with the adversities that the business was facing and so it was a challenge for us. But eventually with constructive meetings and strategy planning despite much difficulties, we paved out a way to keep the business running. One of my brothers proposed the delivery system maintaining proper hygiene and safe distance. We started implementing the plans by approaching our loyal customers. Eventually, we kept receiving orders from customers. Along with my other two brothers, we served doorstep delivery maintaining proper hygiene and sanitization. So as a result, in one line, the business did not stop entirely due to lockdown.

Question: How has your business been post-lockdown and what sorts of products are you selling presently?

Abhishek Nandi: As I said earlier, that gift items and decorative items were the products kept in stock during the infant stages of the business. Eventually, the business grew and evolved. So currently, we primarily have two broad categories: stationery items and grocery items. For example, grocery items include variety of biscuits, chips, soft drinks, packaged water bottles, etc. And in case of stationery items: sorts of pens, pencils, notebooks, and few more office items. So these are the kind of items available at our store. Plus-minus, our business is running at a considerable pace.

Problem Statement

- Maximizing the existing revenue generation. On a glimpse of the data received for one month, it is found that for a few

categorical items, there is a possibility for increasing the revenue.

- Since, there exists a stabilized reputation of the shop for its services, we will explore the possibilities of introducing a new product which will increase the profit of the business.

Metadata and Descriptive Statistics

All the metadata and descriptive statistics is as per the given link of the analysis and the raw data.

The sheet named “Daily Sales (Raw data)” is the modified cleaned spreadsheet which was received from the owner of the shop, Mr. Nandi. The sheet shows the data for a period of four months. It contains the daily sales data. The columns store the sales record and the rows contain the dates of the sales. Column heads show the name of the product being sold with its respective category to which it belongs. There are 29 products belonging to 9 different categories. This sheet is the primary source of data on which the entire analysis is based.

The sheet named “Rate Chart” displays the price of each product along with its categories.

The sheet named “Categorical Daily Sales” displays the revenue generated by each product of each category on a daily basis.

The next sheet named “Pivot Table” is the sheet containing the pivot table created using the source data in “Daily Sales (Raw Data)” to easily get the summary data for different results and findings. This is done using MS Excel tools.

Detailed explanation of Analysis Process/Method

The Analysis of the source data has been done in the sheet named “Miscellaneous Analysis”. All analysis has been done using the available tools in MS Excel.

Using the pivot table, different kinds of tables have been created for the purpose of analysis.

Referring to the pivot table, the categorical revenue table has been extracted. The rows have been set to “Category” and the values have been set to “Revenue”. With the help of this extracted table, a bar chart has been created to graphically view the levels of revenue for each category. The categorical revenue chart has been created by selecting the categorical revenue table followed by selecting the option “Column Chart” from the tool named “chart” available in the “insert” tab.

Categories	Revenue
Beverages	₹681,280.00
Biscuits	₹179,316.00
Cosmetics	₹582,815.00
Snacks	₹1,243,449.00
Writing Media (Eraser)	₹57,456.00
Writing Media (File)	₹213,324.00
Writing Media (notebooks)	₹426,575.00
Writing Media (Pen)	₹91,080.00
Writing Media (Pencil)	₹496,932.00
Total	₹3,972,227.00

Referring to the previous table, the top 5 revenue categories were filtered out and extracted in a separate table.

Top 5 revenue generating categories	
Snacks	₹1,243,449.00
Beverages	₹681,280.00
Cosmetics	₹582,815.00
Writing Media (Pencil)	₹496,932.00
Writing Media (notebooks)	₹426,575.00

Based on the already created pivot table, the monthly sales for each category along with the daily average sales were calculated. The rows have been set to “Category” and the values have been set to “Sales” and the columns were set to “Months”. The total sales have been calculated using the figures available in the entries of this extracted table. Using the column of “Total Sales”, the daily average sales of each category has been calculated. The formula for the daily average sales is total sales for the category divided by total number of days available in the data. Using the column of “Daily Average Sales”, a pie chart has been created to get an overview of the percentage of contribution in the daily sales for each category. The pie chart has been created by selecting the column of “Daily Average Sales” followed by selecting the option “Pie Chart” from the tool named “Chart” available in the “insert” tab.

Category	August	November	October	September	Total Sales	Daily Average Sales
<i>Beverages</i>	4444	4291	4640	4370	17745	145
<i>Biscuits</i>	1362	1386	1292	1338	5378	44
<i>Cosmetics</i>	1805	1902	1986	1700	7393	61
<i>Snacks</i>	7150	6547	6766	6605	27068	222
<i>Writing Media (Eraser)</i>	373	397	437	389	1596	13
<i>Writing Media (File)</i>	2586	2416	2510	2487	9999	82
<i>Writing Media (notebooks)</i>	3408	3267	3376	3325	13376	110
<i>Writing Media (Pen)</i>	2389	2342	2315	2062	9108	75
<i>Writing Media (Pencil)</i>	4465	4106	4363	4084	17018	139
Grand Total	27982	26654	27685	26360	108681	

Also, selecting the “Total Sales” column, a column chart is created by selecting the tool named “Chart” from the “insert” tab in the same way as that of categorical revenue chart. The chart will facilitate the analysis as it gives a graphical summary of the total sales for each category.

From the categorical monthly sales table, the top five selling products have been extracted from the categorical monthly sales chart.

Next, using the pivot table, the amount of revenue generated day wise is extracted. From the pivot table, the rows are set to “Day” (days of the week) and the values are set to “Revenue” summarized as sum of revenues. Now from this available table, the average revenue generated per day is calculated.

Day	SUM of Revenue	Average Revenue
Mon	593048	4861
Tue	578951	4746
Wed	576214	4723
Thu	551994	4525
Fri	535520	4390
Sat	578115	4739
Sun	558385	4577
Grand Total	3972227	

Besides having the table of values, a bar chart is also created using the tools of MS Excel as mentioned earlier. The bar chart will help us clearly analyze the trend of revenue generation over the entire period based on the days of the week. To get a daily average revenue generation based on the days of the week, a separate table has also been extracted using the pivot table in a similar fashion.

Next, using the same pivot table, the categorical monthly sales and revenue table has been extracted. In both the cases, the rows have been set to “Category”, columns set to “Months”. The only difference lies in the values which are set to “Sales” in case of the sales data, and “Revenue” in case of revenue data.

Using these two tables, a line chart has been created to analyze the flow of revenue generation and sales output for each category over the period of these four months that is from August to November. It basically displays the trend-line that is followed by the category over the entire period. These two graphical charts are named as “Categorical Sales Trend” and “Categorical Revenue Trend”.

Results and Findings

No proper conclusions have been drawn yet. Just the data and the foundation for the analysis has been done using the tools of the software MS Excel. The entire is yet to be properly examined and analyzed to reach a conclusion or a solution for fulfilling the aimed objectives. Hence all of these will be looked into in the coming days and accordingly get reported.